

ASSESSMENT FOR LEARNING

(Text-book based on the New Syllabus
introduced in 2021 by the
Tamilnadu Teacher Education University, Chennai)

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PREFACE

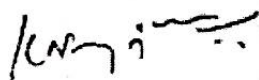
This text book titled **“ASSESSMENT FOR LEARNING”** has been written based on the new Syllabus of the B.Ed. Course of the Tamil Nadu Teacher Education University introduced in 2021. The five units of this subject has been discussed in detail under the following headings:

1. Basics of Assessment
2. Assessment for Learning in Classroom
3. Tools and Techniques for Classroom Assessment and Issues
4. Assessment Practices in Inclusive School
5. Prevalent Practices of Assessment & Reporting of Quantitative Data

To facilitate the easy understanding of the subject by the teacher trainees, the contents have been presented under different headings and sub-headings. As of our previous works, this book is also written in a lucid style with vivid explanations and apt examples. To help students prepare for their examinations, review questions and the expected answers for the same have been provided at the end of each Unit. A set of Model Question Papers based on the University Examination pattern with the expected answers for the same cited in the book, have been provided with this book to help students who are to take their B.Ed. Degree Examination under the new syllabus for the first time.

I am thankful to Mr.C.R.Manivasagan, M.Com., M.Ed., former Office Superintendent of Meston College of Education, Chennai – 14 for his pains-taking efforts in proof reading this book, and Mr. C. Boopalan of Printech India for neatly printing this book.

Constructive suggestions are quite welcome to refine this book further. I earnestly solicit the continued support of teacher-educators and student-teachers, as they have been offering for my previous works. It will be highly satisfying to me if this book proves useful in the hands of student community.



(Dr.K. Nagarajan)

Chennai

04.03.2022

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1. Symbols (A), (B), (C) used in Review Questions refer to Essay, Short Answer and Very Short Answer type questions respectively.
2. Essay type questions are to be answered in about 5 pages.
3. Short Answer type questions are to be answered in about 2½ page.

UNIT – I

BASICS OF ASSESSMENT

1:00 Introduction

It is necessary to assess now and then, the progress achieved in students' learning as a result of their self efforts as well as teacher's classroom teaching. This is an important duty of the teacher too. The term 'Evaluation' is more appropriate to refer this than the term 'Assessment'. Only on evaluating students' learning progress, teacher could know how far his students have understood the subject content taught based on which he could proceed with his teaching further.

This unit deals with the meaning and definition of the terms 'Measurement', 'Assessment' and 'Evaluation', role of assessment in learning i.e. the three approaches of assessment viz. assessment as learning, for learning and of learning, role of teacher in each of these three approaches, formative and summative assessments, their uses, differences between formative and summative assessments, purposes of assessment, principles related to different stages of an assessment programme i.e. principles related to the selection of methods for assessment, collection of assessment information, judging and scoring of student performance, summarization and interpretation of the obtained results and reporting of assessment findings.

1:01 Measurement

1:01:1 Meaning of 'Measurement'

It is quite obvious that 'measurement' is involved in all aspects of human life, in one way or other. Measurement is used to assess the physical characteristics and man's external features. Measurements are used to find, students' height, weight, age, intelligence, their abilities expressed in different fields etc.

The element that is involved in measurement may be any quality or attribute like height, weight, age, intelligence, speed, volume, length, width, etc. For expressing how much a particular quality or attribute is present in an object, person or event, a numerical value is assigned.

Examples :

Krishnan's height is 165 c.m.

Murugan's weight is 60 k.g.

Meenakshi has scored 65 marks in the mathematics test.

In the above examples 165 c.m., 60 k.g., 65 marks are all 'measurements'. Thus assigning numerical values to represent the amount of an attribute or quality present in an object, person or event constitutes 'measurement'.

1:01:2 Definition of 'Measurement'

James M. Broadfield defines measurement as "Assigning numerical values to objects, persons or

events, according to certain well established rules so as to represent the amount of the qualities or attributes present in them." Analysis of the above definition reveals the following :

- i) We can not measure any object, person or event; only the attributes present in them could be measured. For example, a table can not be measured; but only its height, weight, length, width etc. which are called 'attributes' could alone be measured.
- ii) The process of measurement involves, assigning numerical value to represent how much a particular attribute is present in the object, person or event.
- iii) Previously established rules are to be followed for assigning numerical values to represent how much a particular attribute is present.

Stated briefly, measurement in a field involves the following three steps :

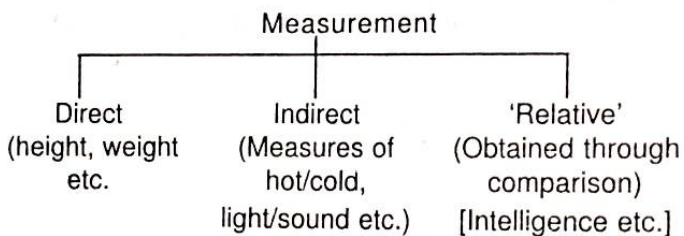
- a) Identifying or defining the attribute or quality that is to be measured [(e.g.) height of a person, weight of an object etc.]
- b) Performing a set of operations by which the attribute may be made manifest and perceivable. [For example, if the height of an individual is to be measured, he or she should be made to stand erect without bending and marking the distance between the head to toe on a stick or thread.]

- c) To quantify the distance marked, an appropriate numerical value is to be given in foot and inch or meter and centimeter (these are called scales based on well established rules) i.e. the height of the individual is to be represented as 5 ft. and 5 inches or 165 centimeters.

1:01:3 Types of Measurement

External attributes or characteristics could be directly measured with high precision, using various tools and devices. For example, height / length (in metres) volume (in litre or cubic centimeters), weight (kilograms or pounds) could be measured directly using appropriate tools / devices. This type of measurement is known as 'Direct measurement'. However heat, light, sound etc. can not be measured directly; but measured only through the effects they produce. Such kinds of measurement are called 'Indirect measurement'.

In education and psychology, measurement is very complex; direct measurement is not possible. Student's achievement, intelligence etc. are measured only by comparing them with those of his peers. These are called '**Relative Measures**'.



Measurements related to external attributes of objects are accurate and highly stable. The scale of measurement starts from zero and contains units of equal size. Such a scale of measurement is called 'Ratio Scale'. [(e.g.) CGS (Centigrade, Gram and Second) System]. Measurements obtained using 'Ratio Scales' could be added, subtracted, multiplied and divided. For examples a measure of 30 c.m. is twice as that of 15 c.m.; in a cloth of length 40 c.m. if 10 c.m. is removed, then the balance left over in the cloth is of length 30 c.m.

However, measurements obtained in behavioural sciences like education and psychology, are not very accurate; they are relative only. Further, as the location of absolute zero point is not known, the units of the scales used are not of equal size. Such scales of measurement are called '**Interval Scales**'. Measurements obtained in an interval scale could not be added, subtracted, multiplied and divided, among themselves. For example, a student who obtained 60 marks could not be said to be twice proficient as that of a student with a score of 30. Similarly, the marks obtained by a student in different subjects could not be added or subtracted. It is clear that neither the marks obtained by a student in different subjects of study nor the marks obtained in the same subject by different students studying in different systems of education (CBSE, State Board etc.) could be compared.

1:01:4 Characteristics of Educational Measurement

The important characteristics of educational measurement are :

i) No Definite 'Zero' Measure

In a mathematics achievement test if a student scores zero mark, it does not mean he has no knowledge in mathematics; it simply means, in that particular test he has not scored any mark. If a different achievement test in mathematics is administered, he may score 35 marks. Thus measurements in education are not precise and accurate; they are simply relative measures.

ii) Unit are not of equal interval

The proficiency of a student scoring 80 marks in an achievement test, could not be taken as twice that of another student with a score of 40 marks in that test. A particular student may get different scores in different tests. Similarly different students get different marks in a particular test. Therefore the marks obtained by a student in different subjects of study as well as the marks obtained by the students in the same subject, studying different sections (of the same grade) could not be compared.

iii) Tests in education measure only the expressed traits or attributes

An educational test does not measure the true potential of a trait in an individual; it measures only

the expressed level of that trait i.e. there is always a difference between the '**potential level**' and '**expressed level**' of a trait present in an individual.

- iv) In education, attributes are not directly measured; only the effects of an attribute on behaviour are measured. For example, the efficiency of memory of a student is measured by making him recall the list of words he was shown once.
- v) Educational measurement is a means and not an end in itself.

For example, if a student scores 67% in a mathematics test, it is not his final goal; it reveals how far he is from the goal to be achieved fully. This serves as a stimulant to take further efforts to improve his performance.

1:02 Assessment

1:02:1 Meaning of 'Assessment'

Attributes / traits like singing, dancing, beauty, honesty etc. cannot be measured accurately. We could assess these qualities by using tools and techniques like observation and tests. Such kinds of quality judgement is known as 'assessment'. For example, the revenue inspector assesses the present value of a house on the basis of the size of the house, the various materials used in the construction of the house and their quality, such as teak / countrywood used for doors and windows, tiles / marbles for the floor, provision of

lifts etc. Based on the present value of the house, he determines the annual house tax to be collected.

Putting it in brief, assessment is the process of objectively understanding the state or condition of a thing, by observation, measurement and tests.

Assessment of teaching means measuring its effectiveness. 'Formative' assessment is measurement for the purpose of improving it. 'Summative' assessment is what we call normally 'evaluation'.

1:02:2 Definition of Assessment

Assessment is defined as **"objectively determining the present state or condition of a thing or process, by observation, measurement and tests"**. The purpose of assessment is to improve the present state or condition of a thing or process.

1:03 Evaluation

1:03:1 Meaning of 'Evaluation'

Evaluation is the process of observing, measuring and testing a thing for the purpose of judging it and of determining its 'value' either by comparison to similar things, or to a standard. Evaluation of teaching means passing judgement on it by observing teaching of a teacher, comparing it with the teaching of other teachers or with some standards / criteria.

Stated briefly, evaluation of a thing or process involves the assessment of its different aspects or features and based on that determining its value.

Therefore evaluation is more broader and comprehensive than assessment. Evaluation includes assessment. 'Measurement' is a tool or method employed in assessment and evaluation.

1:03:2 Definition of Evaluation

Evaluation is the process of determining the value of a thing or process, based on the assessment of its different aspects / components by employing observation, measurement and tests.

Evaluation includes the process of 'assessment', and the tools and methods of 'measurement'.

1:03:3 Illustrations for Measurement, Assessment and Evaluation

- i) Teacher measures the height of Rahul as 125 c.m. and judges that he is the shortest in the class (Teacher evaluates).
- ii) Teacher measures the proficiency achieved by Hemalatha in Mathematics as 65% and judges her performance as "Satisfactory" (Teacher evaluates)
- iii) Mr. Jagadeesh measures the size of the room as 4 m x 3 m x 2.5 m and finds that there are no windows in it. He assesses that the room is too small for 40 students to sit and there is no free air circulation also. Based on his assessment of the room, he submits report to the headmaster that the room is unfit for classroom instruction. (Teacher evaluates).

- iv) Kanthan and Kumaran studying in the same class, if score 85 and 35 marks respectively in the half-yearly examination in mathematics, then Kanthan may be evaluated as more proficient than Kumaran by the teacher (Teacher evaluates).

From the above illustrations, the equation expressing the relationship between assessment and evaluation could be obtained as follows :

$$\begin{array}{l} \text{Evaluation} \\ \text{of} \\ \text{a thing} \end{array} = \begin{array}{l} \text{Assessing the} \\ \text{thing through} \\ \text{observation and} \\ \text{measurement} \end{array} + \begin{array}{l} \text{Determining the value of the} \\ \text{thing by comparing it} \\ \text{with other similar} \\ \text{things or to a standard} \end{array}$$

1:04 Role of Assessment in Learning

Assessment plays three kinds of roles in learning. They are :

- i) Assessment as Learning
- ii) Assessment for Learning
- iii) Assessment of Learning

1:04:1 Assessment as Learning

As students are able to judge about themselves and their learning by the process of assessment i.e. they become aware of what kind of learners they are and how they are learning, it enables them to adjust and advance their learning. Since assessment of learning helps students to know about their strengths and weaknesses, monitor their own learning and improve further, they become metacognitive. Assessment as Learning, develops in students the

knowledge and skills required to adapt themselves to the learning situation and bring improvements in their learning.

Students reflect on their work on a regular basis usually through self and peer assessment and decide on the content areas which require more attention and focus. In short, **assessment as learning helps students to take more responsibility for their learning and fix personal goals for further learning.**

Putting it in a nut shell, **when the student uses assessment in learning to learn further and bring improvements upon it, then the assessment becomes a learning.**

1:04:1:01 Role of the Teacher in Assessment as Learning

In assessment as learning the teacher serves as a guide to students. Students learn self-monitoring their work and make necessary adaptations.

Teachers can adopt the following classroom practices in practising assessment as learning.

- i) model and teach the skills of self-assessment.
- ii) discuss with the students the learning outcomes of each content area in the curriculum.
- iii) provide examples and models of good practice and quality work that reflect curriculum outcomes.
- iv) work with the students and develop the criteria for the various tasks that are to be completed and the skills that need to be learned.

- v) provide regular feedback to students as they learn and by putting guiding questions help them monitor their own learning.
- vi) guide students in setting their own goals, and monitoring their progress towards them.
- vii) create an environment where it is safe for students to take chances and where support is readily available.

Once students acquire meta cognitive skills they become independent learners, adjust their learning as and when required and demonstrate the qualities of self-reflection, self-monitoring and self-adjustment. Following questions help students to acquire meta cognitive skills :

- 1) What is the purpose of learning these concepts and skills?
- 2) What do I know about this topic?
- 3) What strategies do I know that will help me learn this?
- 4) Am I understanding these concepts?
- 5) What are the criteria for improving my work?
- 6) Have I accomplished the goals I set for myself?

1:04:2 Assessment for Learning

'Assessment for learning' is defined as the process of seeking and interpreting evidence for use, by learners and their teachers to decide where

the learners are in their learning, where they need to go and how best to get there (Assessment Reform Group, 2002).

The emphasis shifts from summative to formative assessment in 'Assessment for Learning'. Assessment for learning happens, when a teacher monitors students' progress in learning during the teaching of a content unit, identifies the shortcomings in their learning, conducts tests intermittently during the period of instruction and provides continuous feedback to students with a view to remove gaps in students' learning. The continuous feed back obtained from such formative assessments of students helps the teacher to plan and incorporate necessary modifications in his teaching. Similarly students too make use of the feed back provided to them to remove the shortcomings in their learning, try to continuously improve and refine it.

Summing up it could be said, assessment for learning is the process of improving and enhancing students' learning. Such an assessment is undertaken throughout the period of teaching, helping both students and the teacher to know the progress and shortcomings in students' learning by providing continuous feedback. The aim of assessment for learning is to bring improvement in their present learning.

Stated briefly, in assessment for learning, teachers use assessment as an investigating tool to find out as much as they can about what their

students know and can do, and what confusions, preconceptions, or gaps they might have. It provides the basis for providing descriptive feedback for students to correct their shortcomings in learning. Thus assessment for learning includes both diagnostic and formative assessments.

1:04:2:01 Teachers' Role in Assessment for Learning

Teachers' role in assessment for learning i.e. assessment for improving students' learning is as follows:

- 1) Assessment for learning that happens during the entire period of teaching will be in the form of teachers' interactions with students i.e. it may be in the form of questions put intermittently during teaching or it may be the home assignment given to students testing whether they have understood what have been taught in the class or may be other assessment activities meant for improving students' learning.
- 2) Identifying the particular learning needs of students and the learning deficiencies of backward students, from the feedback obtained from the assessment.
- 3) Selecting and adopting materials and resources that help to improve students' learning.

- 4) Developing differentiated teaching strategies and learning opportunities for helping backward students in learning.
- 5) Providing immediate feedback and direction to students to improve their learning.
- 6) Aligning instruction to suit the needs of students.

1:04:3 Assessment of Learning

Assessment of learning is the use of a task or an activity to measure, record and report on a student's level of achievement with respect to specific learning expectations. These are often known as summative assessments.

It is usually held at defined points during a course of study, say at the end of a unit, term or semester to collect evidences how far the instructional objectives are achieved in terms of students' learning achievement. Such assessments are used to rank or grade students. Assessment of students' learning achievement helps in deciding who are all to be promoted to the next higher class or permitted to go to the next content unit.

1:04:3:01 Teachers' Role in Assessment of Learning

Teachers' role in assessment of learning is listed as follows :

- i) Reporting to parents and making it public, the level of learning achievement of students, with evidence obtained by assessing the different aspects of learning.
- ii) Providing a rationale for undertaking a particular assessment of learning.
- iii) Assessing students' learning achievement in relation to the expected learning outcomes i.e. assessment of students' learning proficiency in relation to the instructional objectives.
- iv) Making use of different kinds of test items (like fill up the blanks, true/false, multiple choice, matching the items in the two columns) and techniques (like written tests, oral tests, practical examination etc.) in the assessment of students' learning.
- v) Interpreting and grading students' learning achievement in relation to the acceptable level of learning achievement.
- vi) Adopting transparent approaches in interpreting students' marks.
- vii) Being ready with strategies for resolution in the event of disagreement regarding students' learning achievement.

1:04:4 Comparative Analysis of Three Kinds of Learning Assessment

S. No.	Bases for the Analysis	Assessment as Learning	Assessment for Learning	Assessment of Learning
1.	Purpose of Assessment	To guide and provide opportunities for each student to monitor and critically reflect on his or her learning and identify next steps.	To enable teachers to determine next steps in advancing student learning.	To certify or inform parents or others of students' proficiency in relation to curriculum learning outcomes.
2.	What are Assessed?	Each student's thinking about his or her learning, the strategies he or she uses to support or challenge that learning and the mechanisms she or he uses to adjust and advance her or his learning.	Each student's progress and learning needs in relation to the curricular outcomes.	The extent to which student can apply the key concepts, knowledge, skills and attitudes related to the curricular outcomes.
3.	Assessing Methods	A range of methods in different modes that elicit students' learning and metacognitive processes	A range of methods in different modes that make students' skills and understanding visible.	A range of methods in different modes that assess both product and process.

S. No.	Bases for the Analysis	Assessment as Learning	Assessment for Learning	Assessment of Learning
4.	Ensuring Quality of Assessment	+ Accuracy and consistency of student's self-reflection, self-monitoring and self-adjustment. + Engagement of the student in considering and challenging his or her thinking. + Students record their own learning.	+ Accuracy and consistency of observations and interpretations of student learning. + Clear, detailed learning expectations. + Accurate, detailed notes for descriptive feedback to each student.	+ Accuracy, consistency and fairness of judgement based on high quality information. + Clear, detailed learning expectations. + Fair and accurate summative reporting.
5.	Using the information obtained from assessment	+ Provide each student with accurate descriptive feedback that will help him or her develop independent learning habits. + Make each student focus on the tasks that are to be learnt and his/her acts of learning.	+ Provide each student with accurate descriptive feedback to further his or her learning. + Differentiative instruction by continually checking where each student is in relation to the curricular outcomes.	+ Indicate each student's level of learning. + Provide the foundation for grading students on the basis of their learning proficiency and deciding who are all to be promoted to the next higher class.

S. No.	Bases for the Analysis	Assessment as Learning	Assessment for Learning	Assessment of Learning
		+ Provide each student with ideas for rethinking, adjusting and articulating his / her learning. + Provide the conditions for the teacher and student to discuss alternatives. + Students report about their learning.	+ Provide parents or guardians with descriptive feedback about student learning and ideas for lending support to the learning of their children.	+ Report fair, accurate and detailed information that can be used to decide the next steps in a student learning.
6.	Type of Assessment	It is formative; promotes self-learning and helps the learner to improve his / her learning.	It is formative and diagnostic; helps the teacher to find the learning needs of students and modifying his teaching accordingly.	It is summative; it is highly useful for parents, teachers and educational administrators.

1:05 Formative and Summative Assessments

Measuring students' behavioural changes based on well defined educational objectives formulated and interpreting the obtained scores meaningfully, find an important place in educational assessment. Such assessments are of two types - (i) Formative assessment and (ii) Summative assessment.

1:05:1 Meaning and Definition of 'Formative Assessment'

In any event, assessments made at different stages of its growth is called formative assessment. Questions asked during the classroom instruction, assignments given by the teacher everyday at the end of his teaching to assess how far students have understood what have been taught in the class, unit test conducted after a content unit has been taught, weekly and monthly tests conducted by the teacher, are all formative type of assessment.

N.E. Gronlund defines formative assessment as **"the evaluative technique used to monitor learning progress during instruction and to provide continuous feedback to both pupil and teacher concerning learning success and failures."**

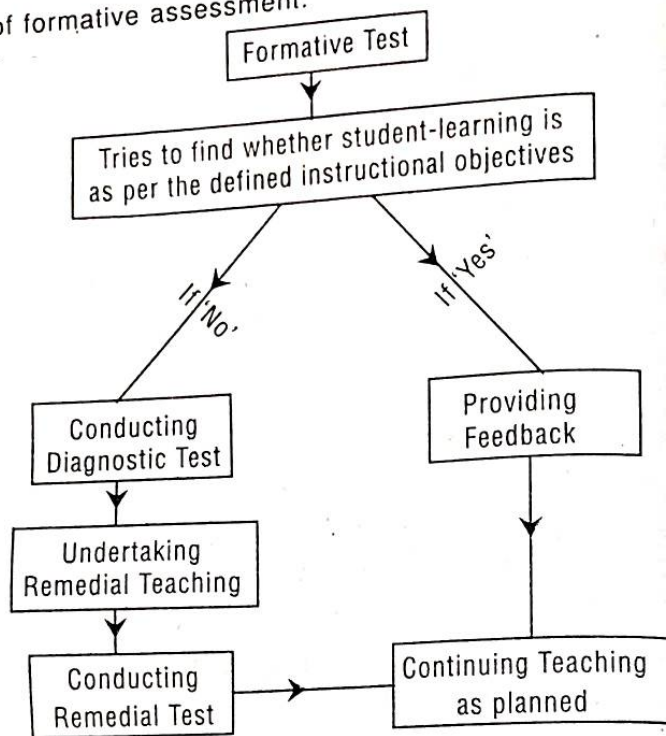
1:05:1:01 Important Features of Formative Assessment

From the definition of formative assessment, its important features could be derived as follows :

- i) It is used for monitoring students' learning progress during the period of teaching.
- ii) It provides continuous feedback to students now and then, regarding their learning progress and thus helps them to rectify their mistakes and shortcomings, paving the way for achieving proficiency in learning.
- iii) It provides feed back to teachers so that they could make necessary changes in their teaching i.e. improving further, the learning experiences provided to students.
- iv) It may be in the form of questions put to students during classroom instruction, home assignments given to students everyday to test how far students have understood what have been taught in the class that day, weekly test or monthly test.
- v) Formative tests in any subject of study are usually developed by the teacher who has taught it to the students.
- vi) It is used to test, how well students learn what are being taught, as the process of teaching is going on.
- vii) It helps the teacher to align his teaching to suit the immediate needs of the learners by adopting necessary changes in his instructional methods and techniques.

Stated briefly formative assessment aims to improve student learning and is not useful in grading students on the basis of their learning achievement.

The following diagram throws light on the nature of formative assessment.



1:05:1:02 Uses of Formative Assessment

Formative assessment conducted during teaching is used for the following :

- To find how far the instructional objectives are achieved?
- When instructional objectives are not achieved, formative assessment helps to conduct diagnostic test and undertake remedial teaching for those who lag behind in learning.

- Formative assessment helps the teacher to change the method and techniques of his teaching, based on student responses during instruction.
- Formative assessment helps in sustaining students' motivation in learning.
- Formative assessment reduces students' test anxiety.
- Formative assessment helps to inform the parents now and then about the learning progress of their wards, enabling them to take necessary steps to improve their learning.

1:05:2 Summative Assessment

1:05:2:01 Meaning and Definition of 'Summative Assessment'

Assessment carried out at the end of any event, activity or process is known as 'summative assessment'. In education, assessment of the total output at the end of any educational or training programme is called summative assessment.

According to Wierman and S.G. Gurs "Summative evaluation is done at the conclusion of instruction and measures the extent to which students have attained the desired outcomes".

1:05:2:02 Important Features of Summative Assessment

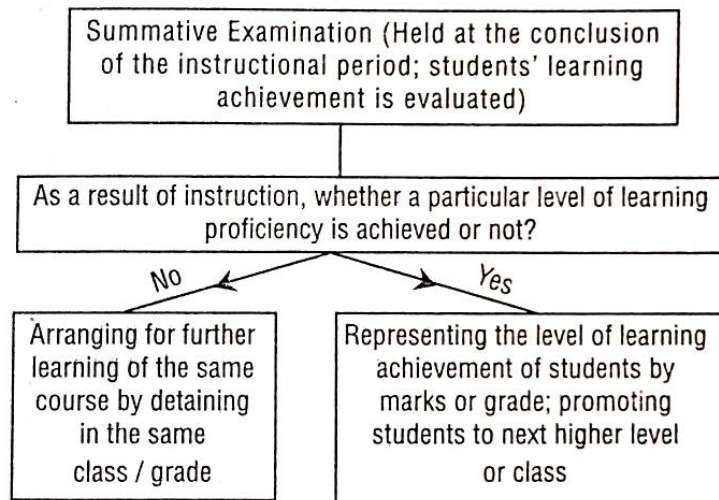
The following are the main features of summative assessment.

- i) It is generally carried out at the end of a particular instructional period, which may vary from one semester to one full academic year. Test used to assess the proficiency achieved by students at the end of the instructional period is known by the name 'examination'. In Indian educational institutions, conducting quarterly examination, half-yearly examination, annual examination, semester examination etc. are in vogue at present.
- ii) Summative assessment attempts to find the proficiency achieved by the students against the instructional objectives established for course of study or educational programme.
- iii) In summative assessment i.e. in terminal examinations, learning proficiency achieved by students in different parts and aspects of the syllabus for the course, is evaluated and represented by marks or grades.
- iv) Summative assessment could be used to grade students on the basis of their level of learning achievement. It is also often used to decide who are all among the students, eligible for promotion to the next higher class or allowed to go to the next unit of the subject content.
- v) Many kinds of test items, drawn from the different portions of the subject content find place in summative tests / examinations.

- vi) Test items in summative assessment vary in the difficulty level, from easy to difficult and very difficult. Similarly test items may evaluate students' learning in the cognitive domain at the three levels i.e. knowledge, understanding and application. Skill-learning will be tested through 'Practical Examination'.

Putting it in short, summative assessment includes public examination, terminal examination, annual examination etc.; It may be a teacher-made test or rating scale. Summative assessment helps evaluate the instructional programme on the whole, through the learning achievement of students.

The following diagram illustrates the main features of summative assessment.



1:05:2:03 Important Uses of Summative Assessment

- i) To identify the present level of educational proficiency of students
- ii) To classify students on the basis of their learning achievement
- iii) To take the decision whether or not, to promote the student to the next higher class.
- iv) To decide the further course of the educational journey of students, based on the marks or grades obtained.
- v) To find the overall performance of the teaching - learning process.

1:05:2:04 Differences Between Formative and Summative Assessment

The following table explains the differences between formative and summative assessments :

S. No.	Formative Assessment	Summative Assessment
1.	Is done using tests.	Is carried out with Examinations.
2.	Is carried out during the period of instruction, at each stage.	Is done at the conclusion of the instructional period or academic session, evaluating the overall achievement of students.
3.	Content area being tested is small; but indepth evaluation is done.	Tests large portion of the content taught, evaluating the different aspects of learning; however evaluation lacks depth.

S. No.	Formative Assessment	Summative Assessment
4.	Leads to diagnostic testing and remedial teaching for those who are found to lag behind in learning.	Helps to grade students based on achievement level and decide who are all to be promoted to the next higher class.
5.	Reveals how well students have learned the portion taught.	Assesses the overall achievement of students in each subject of the course of study.
6.	Is an integral part of the teaching - learning process.	Is carried out at the terminal stage of the teaching - learning process.

1:05:2:05 An Illustration from Practical Life for Formative and Summative Assessment

In a factory manufacturing wrist watches, there are different sections to produce each part of the wrist watch. In each section, every piece of the particular part of the watch it produces, will be tested thoroughly. This amounts to formative assessment. All the sections send the parts they prepared to the assembly section, where the parts are assembled to make the watch in its final shape and packed in boxes. Each box is tested and certified for the smooth functioning of watches. This final stage testing is otherwise known as summative assessment. In this terminal stage testing, only one or two pieces of watches are taken from each lot for testing. The lot containing defective watches are not sent for distribution to dealers.

1:06 Purpose of Assessment

The following are the main purposes of educational assessment.

1. Assessment drives instruction

Before start teaching, teacher by using pre-test or need assessment could find what the students already know and use the prior knowledge as a stepping place to develop new knowledge in students through his teaching. The same is true for data obtained through assessment done during instruction. By testing students' learning now and then throughout instruction, the teacher could constantly revise and refine his teaching to meet the diverse needs of students.

2. Assessment drives learning

What and how students learn depends to a major extent on their knowing of how their learning achievement will be assessed. Assessment practices should send the right signals to students about what to study, how to study and the relative time to spend on concepts and skills in a course. Students understand what they need to know and be able to do, both through a clearly articulated syllabus as well the achievement tests conducted and assignments given by the teacher and they prepare themselves accordingly. High expectations for learning, result in

students taking increased efforts and rise to the occasion.

3. Assessment informs students of their progress

Assessment helps students

- i) to know about their present level of progress in learning and how far their efforts have been fruitful.
- ii) to find how is their progress in relation to the learning outcomes of the course.
- iii) to get motivated in learning by getting feed back from assessments now and then regarding their strengths and shortcomings in learning.
- iv) to take remedial measures based on the feedback obtained from assessment.

4. Assessment helps teachers to know the effectiveness of their instructional method

Reflection on student accomplishments offers teachers insights on the effectiveness of their teaching strategies, based on which they can adopt necessary modifications in their teaching so as to enhance its effectiveness.

5. Assessment grade students on the basis of their level of learning achievement

Assessment grade students according to their proficiency attained in learning. Assessment helps to

find whether the level of a student's learning achievement is on par with the level of accepted standard or not and decide accordingly to promote him / her to the next higher class or allow him to take up the next unit of the object content for learning.

6. Helping the teacher to take remedial measures when student learning is not upto the expected level

From the feedback data obtained from assessments, the teacher could understand in which areas or portions of the content taught, students' performance is not satisfactory and accordingly reteach the same and arrange for those backward in learning remedial teaching on the basis of the findings of diagnostic test.

Summing up, it could be stated that based on assessment, students' learning achievement and their further journey in education are determined.

1:07 Principles of Good Assessment Practice

Substantial research exists on the characteristics of good practice of assessing student learning. The findings of the various research studies concerning the characteristics of good learning assessment practice have been summarized in the following set of principles.

1. The primary purpose of assessment is to improve student performance

Good assessment is based on a vision of the kinds of learning we most value for students and how best they might achieve these. Good assessment practice gives more importance for how things are learned rather than what are learned.

2. Assessment should be based on an understanding of how students learn

Learning is a complex process. It entails not only what students know but also what they can do with what they know; besides knowledge and abilities, it involves values, attitudes and habits of mind that affect both academic success and performance beyond the classroom. Assessment should reflect these understandings by employing diverse methods, using them over time, so as to reveal change, growth and increasing degree of integration. Such an approach aims for a more complete and accurate picture of learning.

3. Assessment should be an integral component of course design and not something to add afterwards

When a course or programme is designed, apart from the teaching and learning elements, assessment of learning, the third important element should be stated clearly and explicitly. Then only students can

demonstrate what they have learned and see the results of their efforts.

4. Good assessment provides useful information to report credibly to parents on student learning achievement

A variety of assessment methods provide teachers with evidence of what students know and can do and their particular strengths and weaknesses. Teachers can then report to parents on how far their child has progressed during that year, where they are as compared to the relevant standards and what the student, parents and the teacher need to do to improve the student's performance.

5. Good assessment requires clarity of purpose, goals, standards and criteria

Assessment works best, when it is based on clearly stated purpose and goals of the course, the standards which students are expected to achieve and the criteria against which success is to be measured. Assessment criteria need to be understandable and explicit so that students know what is expected of them from each assessment they undergo. Students' parents and the community should all be able to know the need for assessment and the reasons for selecting each individual form of assessment in its particular context.

6. Good assessment requires a variety of measures

In general, a single instrument will not tell us all we need to know about students' achievement and how it can be improved. Hence, the teacher need to be familiar with a variety of assessment tools and their uses so that he / she can make use of the most appropriate tool / method of assessment according to the kind of information sought.

7. Assessment methods used should be valid, reliable and objective

Assessment instruments and processes which directly measure what we intend to measure should be chosen. A good assessment system is one which provides for moderation between the assessment measures of different teachers as well as objectivity in measurement.

8. Assessment requires attention to outcomes and processes

Information about the learning achievement of students though very important, to know the level of proficiency attained by each student in the class, it is also equally important to know about their learning experiences along the way and the efforts taken to acquire them.

9. **Assessment works best when it is ongoing rather than episodic**

When learning assessment of students is undertaken continuously, it proves to be most effective. If the various learning related activities of students are monitored and assessed over a period of time, continuous learning improvement takes place and the course-goals are achieved; students' learning achievement too rise to the expected level of standard.

10. **Assessment for improved performance involves feedback and reflection**

All assessment methods and process should provide for students to receive feed back on their learning and performance i.e. assessment should serve as a developmental activity enhancing student learning. Assessment should also provide students and teachers with opportunities to reflect on their practice, efforts and learning achievement.

1:08 **Principle Related to Different Stages of an Assessment Programme**

There are five stages, in the selection and use of appropriate methods to assess students' knowledge, skills, attitude etc. and reporting the obtained findings to the stake holders of students' educational progress. These stages are :

- i. Developing / Choosing methods for assessment
- ii. Collecting the assessment information
- iii. Judging and scoring student performance
- iv. Summarizing and interpreting the assessment results.
- v. Reporting the assessment results.

The five stages involved in any assessment programme and the principles related to each one of them are explained in detail as follows:

1:08:1 **Developing / Choosing Methods for Assessment**

Teacher could make use of different methods and techniques to assess students' subject knowledge, skills and attitudes. These include observation, written tests consisting of subject-based questions, oral testing, interview, peer assessment, self-assessment, standardized tests (criterion referenced and norm-referenced), performance tests, portfolio assessment, process and product assessments etc. Using various such methods of assessment, teachers collect information regarding how far students have achieved progress in curricular knowledge, skills, attitudes and the behaviour that are to be learned.

1:08:1:01 Principles Related to the Selection of Assessment Methods

- i) The methods chosen or developed to assess students' knowledge, skills, attitudes and behaviour should have high validity and consistent with the resources that are available i.e. the assessment methods and tools used should yield valid / appropriate measures which do not give room for any wrong interpretation.
- ii) Assessment methods and tools used should give highly reliable measures i.e. if any one, at any time use the assessment method and tool he / she should get the same result.
- iii) Assessment methods should not only be related to the instructional objectives and goals but also be compatible with instructional approaches.
- iv) Assessment methods chosen / developed should give pertinent information enabling parents and teachers to take decisions regarding the education of the student.
- v) For assessing student performance comprehensively and reliably, more than one assessment method is to be used as there are multiple student attributes that affect their performance.

- vi) Assessment methods and tools should be suitable to the students' background and previous experiences.
- vii) Contents and language style used in assessment tests should not be offensive or irritable.
- viii) Assessment tests translated from another language should be compatible with the cultural environment of the testees and provide valid inferences for the intended purpose.

1:08:2 Principles Related to Collecting the Assessment Information

Assessment data regarding students' learning achievement, level of proficiency attained in skills, desirable attitudes developed, learned behaviour and social skills could be collected in a variety of ways like observation, oral test, interview, interest inventory etc. though guidelines for the procedure to be followed in each of these assessment techniques vary. However, whatever may be the assessment technique used, there are certain general principles related to collecting the assessment data. They are :

- i) Students should be told before hand, why assessment information is collected, what kind of information is to be collected and how this information is to

be used i.e. developing a proper mindset in students to face the assessment is necessary.

- ii) The assessment procedure employed to collect the requisite data should suit the purpose and form of the tool used. It should be understood that there is no assessment technique or tool that could be used in all conditions.
- iii) When assessment techniques like observation, check list and rating scale are used, it should be ensured that attributes to be assessed in a short session are small in number and well defined so that observations could be accurate and reliable.
- iv) The directions provided to students while using an assessment technique or tool should be clear and complete as well as appropriate to their age and grade level.
- v) Assessment techniques which involve choosing the correct alternative among those given like true / false, multiple choice etc. should facilitate students to answer all items in the test, without the threat of penalty for choosing the incorrect alternative as response.
- vi) While collecting assessment information, interactions with students like clarifying their

doubts, replying to their queries etc. should be appropriate and consistent.

- vii) While collecting assessment information, if any unexpected interruptions / interferences take place, such happenings should be recorded.
- viii) Written guidelines for alternative procedures to be followed should be available, for collecting assessment information from students with special needs.

1:08:3 Principles Related to Judging and Scoring Student Performance

Principles related to judging and scoring the performance of students and ensuring consistency in the evaluation procedure, are as follows :

- i) Before using an assessment method, a **detailed scoring scheme** is to be prepared to guide the process of judging the quality of performance of students, the product they produce, appropriateness of the attitude and behaviour they express and the correctness of the answers they give.
- ii) Before using an assessment method, students should be informed how their responses and information they give, will be judged and scored.
- iii) While assessing the responses and information provided by students, factors that are not relevant

to the purpose of assessment, should not be allowed to influence evaluation.

- iv) While assessing and scoring the responses and information provided by students, the comments that are recorded regarding the student performance, should be easy to understand and make use of them.
- v) If it is felt that changes or modifications are necessary in the scoring procedure of student responses during the scoring process (such changes should be based only on real problems encountered in scoring the responses), then the modified procedure should be applied to all the responses already scored and rescore them.
- vi) The scheme of appeal if any for revaluation of answer scripts or performance, it should be explained well in advance, at the beginning of the academic year or course of instruction.

1:08:4 Principles Related to Summarizing and Interpreting Results Obtained in Assessment

Summarizing and interpreting the results obtained in the assessment consists of two procedures viz.

- (i) summarizing the comments on students' performance and (ii) assigning grade symbol for each student which indicates the level of his / her performance and the value associated with it.

Summarizing and interpreting the results of assessment should yield accurate information about each student's performance in relation to the instructional objectives.

Principles related to summarizing and interpreting results obtained in the assessment are as follows :

1. A written policy should guide the procedure for summarizing and interpreting the assessment results obtained in a given period of time.
2. How summary of comments regarding students attainment in the assessment and the allotment of grade symbols for the level of attainment are formulated and interpreted, should be explained before hand to students and their parents.
3. The process of deriving summary comments on individual results and grades should be explained in detail. Then only, students could understand the assessment results of their performance.
4. When different kinds of assessment results are combined to prepare summary comments on student's performance, it should be done more cautiously and carefully. As far as possible student's achievement, efforts, participation and other relevant behaviours should be assessed and graded separately, since each of these attributes may require different scale of measurement.

5. Summary comments on student's performance and allotment of grade should be based on more than one assessment result. Then only adequate sampling of learning outcomes is possible, making the evaluation more reliable.
6. When more than one assessment result is used in preparing the summary comments on students' performance and awarding of grades, proper weightage should be given for each of the assessment result used. For example, the weightage to be given for the quarterly, half-yearly and annual examination marks should be determined well in advance and based on that, the marks obtained by individual students in different examinations are combined for every school subject so as to find the level of achievement of each student in each one of the school subjects.
7. The basis for the interpretation of assessment result should be justifiable. For example for all those who are placed in the grade level 'D', we can interpret their performance level as "Inadequate performance; needs improvement" and this interpretation is highly justifiable.
8. While interpreting assessment results, students background and their learning experiences are to be kept in mind.

9. Assessment result that are combined into summary comments and grades should be well preserved separately, ensuring that they do not lose their accuracy. For example, quarterly an half-yearly marks obtained by students in different school subjects are to be preserved along with ensuring that there is no addition, deletion or any kind of modification.
10. Interpretations of assessment results should take into account the limitations in the assessment methods / tools used, problems encountered in collecting the assessment information, judging and scoring it and the limitations in the basis used for interpretation.

1:08:5 Principles Related to Reporting the Assessment Results

Following are the principles related to reporting the assessment findings.

1. The assessment results presented in the form of a written report should mention the period of instruction or semester to which it pertains to i.e. it should clearly state that the report is meant for quarterly, half-yearly, annual or first semester / second semester.

2. Assessment reports should be prepared in accordance with the written policy of the school with regard to reporting of assessment results.
3. Assessment reports should be complete in their descriptions of strengths and shortcomings of students in each of the school subject. Then only, students can build upon their strengths further and focus their attention more on the problem areas and shortcomings so as to improve their learning achievement.
4. Following the presentation of assessment report to students, there should be a meeting between teachers and parents / guardians to discuss the learning progress of students. Students too could be allowed to participate in such meetings.
5. Whether an appeal for reviewing the examination marks will be entertained in parent-teacher meetings, should be made clear in advance, preferably at the beginning of the academic year itself. If appeal for reviewing the examination marks of students is possible according to the school policy, then the proper procedure for the same should be explained to the students and parents.
6. Discussions between teachers and parents pertaining to assessment reports, parents

- demanding corrections / modification in the assessment results etc. should generally be governed by a written policy of the school, which should be consistent with the applicable laws of the State Department of School Education and with basic principles of fairness and human right.
7. Assessment report should contain accurate and complete information regarding student's learning achievement in different aspects of the curriculum and should also be helpful to parents and others who have a concern in the interest of the student, to plan and help in improving his / her performance.
 8. Transfer of assessment information concerning the student performance, from one school to another should be governed by a written policy with stringent provisions for safeguarding confidentiality.

Conclusion

In this unit, meaning and definition of the terms 'Measurement', 'Assessment' and 'Evaluation', role of assessment in learning i.e. the three approaches of assessment viz. assessment as learning, for learning and of learning, role of teacher in each of these approaches, formative and summative assessments, their uses, differences between formative and summative assessments in education, principles of

good assessment practice, principles related to the five stages of an assessment programme i.e. Principles related to (i) choosing / developing the assessment methods for assessment (ii) Collecting the assessment information (iii) Judging and scoring student performance (iv) Summarizing and interpreting assessment results (v) Reporting the assessment results, are discussed in detail.

Review Questions:

1. What do you mean by the term 'Measurement'? (C) [Ans:1:01:1]
2. Give a suitable definition for the term 'Measurement'. (C) [Ans: In 1:01:2, except the last paragraph i.e. till the end of (iii)]
3. Define the term 'Measurement' and state its need and characteristics. (B) [Ans: 1:01:2 in full]
4. What do you mean by the term 'Measurement'? What are its different types? (B) [Ans: 1:01:1 + 1:01:3]
5. Explain the characteristics of educational measurement. (B) [Ans:1:01:4]
6. What is the meaning of the term 'Assessment'? (C) [Ans:1:02:1]

7. Give a definition for the term 'Assessment' (C) [Ans:1:02:2]
8. Explain the concept 'Evaluation' by giving a suitable definition for it. (B) [Ans: 1:03:1 to the end of 1:03:3]
9. Explain the approach 'assessment as learning' and the role of teacher in it. (B) [Ans:1:04:1 + 1:04:1:01]
10. Explain the approach 'assessment for learning' and the role of teacher in it. (B) [Ans: 1:04:2 + 1:04:2:01]
11. Explain the approach 'assessment of learning' and the role of teacher in it. (B) [Ans: 1:04:3 + 1:04:3:01]
12. Explain the three approaches in assessment of learning and the role of teacher in each one of them. (A) [Ans: 1:04:1 + 1:04:1:01 + 1:04:2 + 1:04:2:01 + 1:04:3 + 1:04:3:01]
13. Compare the three kinds of roles assessment plays in student learning. (A) [Ans: 1:04 + 1:04:1 + 1:04:2 + 1:04:3 + 1:04:4]
14. What is meant by 'Formative assessment'? State its important features and uses. (B) [Ans: 1:05:1 + 1:05:1:01 + 1:05:1:02]
15. What is meant by 'Summative assessment'? State its important features and uses. (B) [Ans: 1:05:2:01 + 1:05:2:02 + 1:05:2:03]
16. Explain formative and summative assessments and list the differences between them. (A) [Ans: 1:05:1 + 1:05:1:01 + 1:05:2:01 + 1:05:2:04]